Current Shunt

—/\/\/\— Very small resistance precision resistor often in the mΩ range

↑
Shunt resistor

+ V_{Drop} −

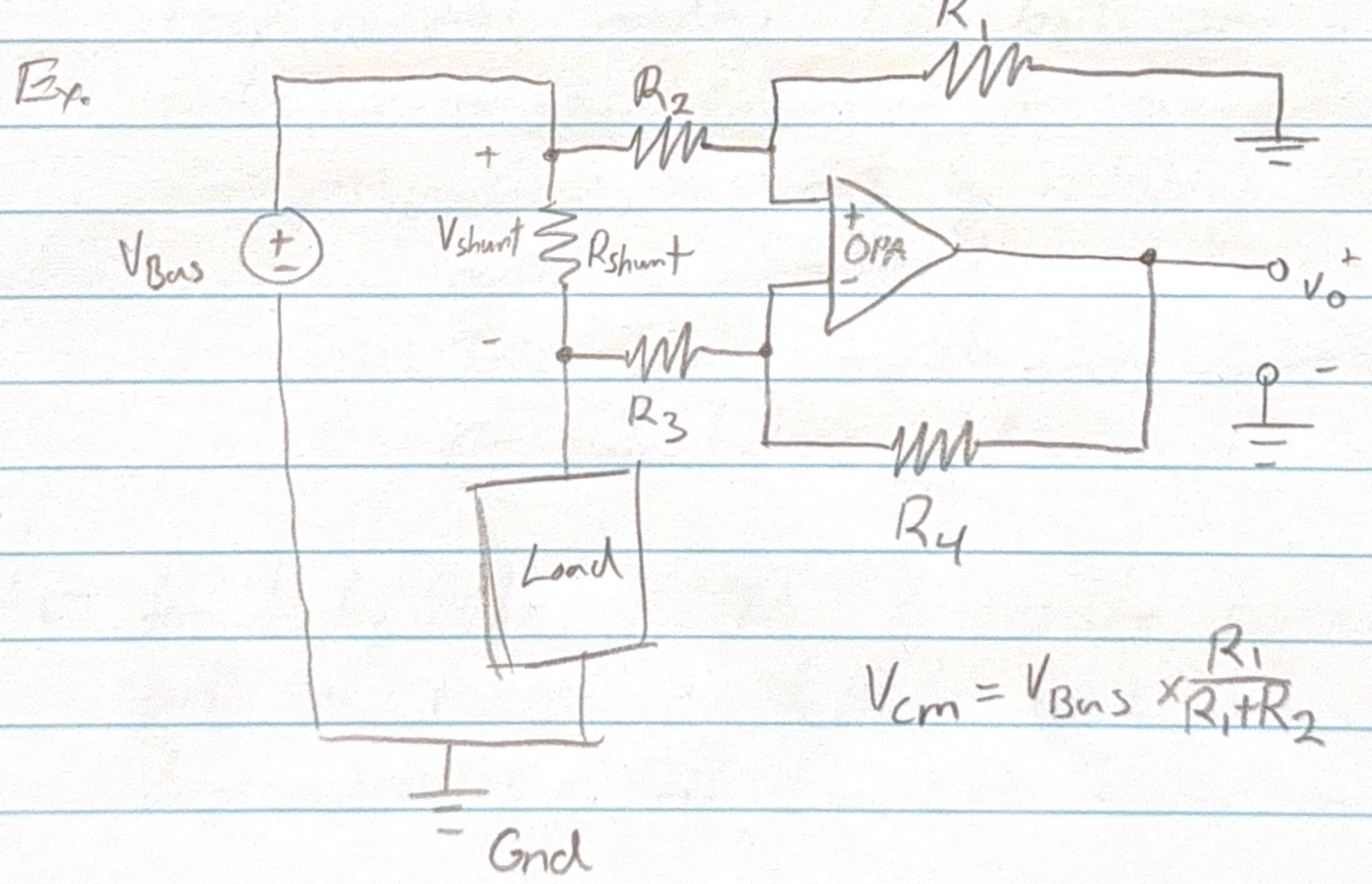
—/\/\/\— Using the extremely small voltage drop across the resistor current can be measured using ohm's law.

$$I = \frac{V_{Drop}}{R}$$

Signal gain

↳ since the voltage drop across the resistor is usually in the ballpark of mV the signal must be boosted to be usable.

↳ An op-amp is wired up to boost the voltage



$$V_{cm} = V_{Bus} \times \frac{R_1}{R_1 + R_2}$$

Problems

- The FS-5 battery has a max voltage of ≈ 108v
  - This means that the op-amp must be rated accordingly
- with a max amp draw of around 600A
  The shunt must be very low resistance and rated for over 600A
- Testing for BSPD
  - A test reading must be done to test the validity of the BSPD
  - This should hopefully be done with less current running through the car.

Possible solutions:

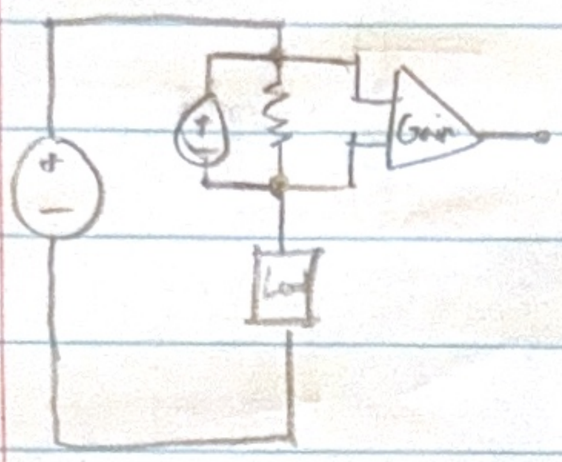
- to decrease the Amps running through the shunts they could be run in parallel to cut the Amprage in half. The measurement would then need to take readings from both shunts and make up for the difference in calculations.
  - pros: Could cut costs by reducing the most expensive component
  - cons: This throws much more complexity into the calculations and measurements.

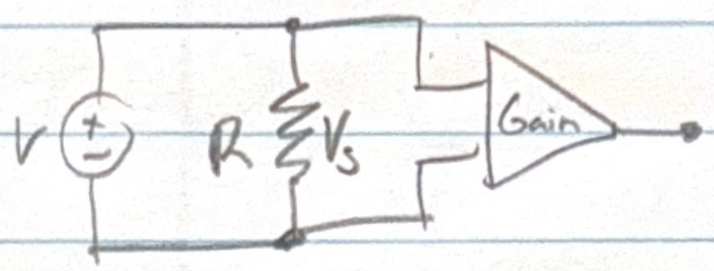
- for testing the BSPD A resistor could be wired with a mosfet switch to read a higher voltage drop across it → this fakes a higher current reading

Possible solutions Cont.

↳ Another way to test the BSPD would be connecting a voltage source in parallel to the shunt. This would create a higher voltage difference.

Bspd voltage boost:

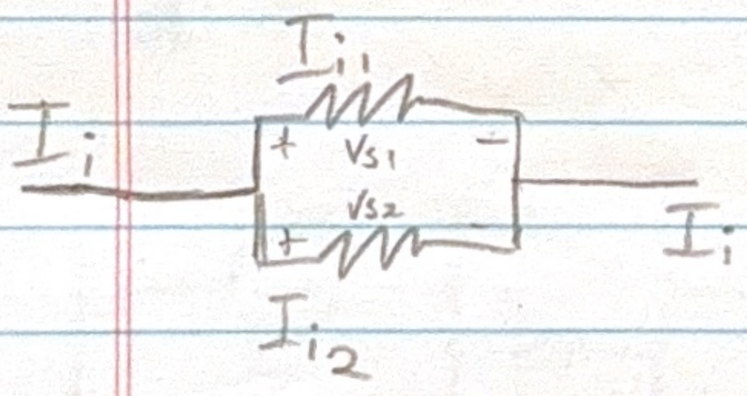




$V_S = V$ → this output means we can artificially boost the current in the shunt resistor to send data to the tractive battery board

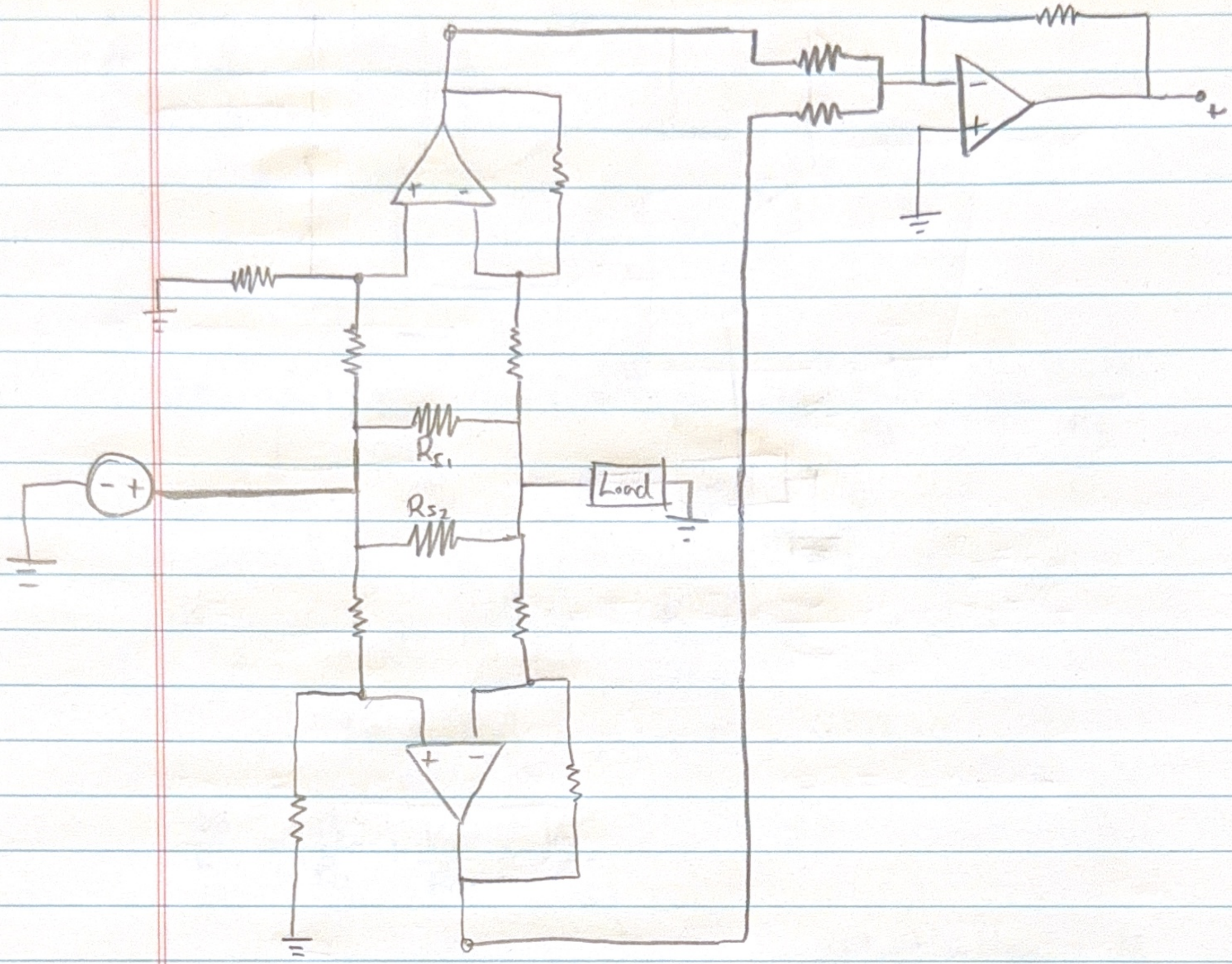
Dual shunts

↳ this circuit will need to contain two Amps to amplify the voltage difference of each shunt respectively

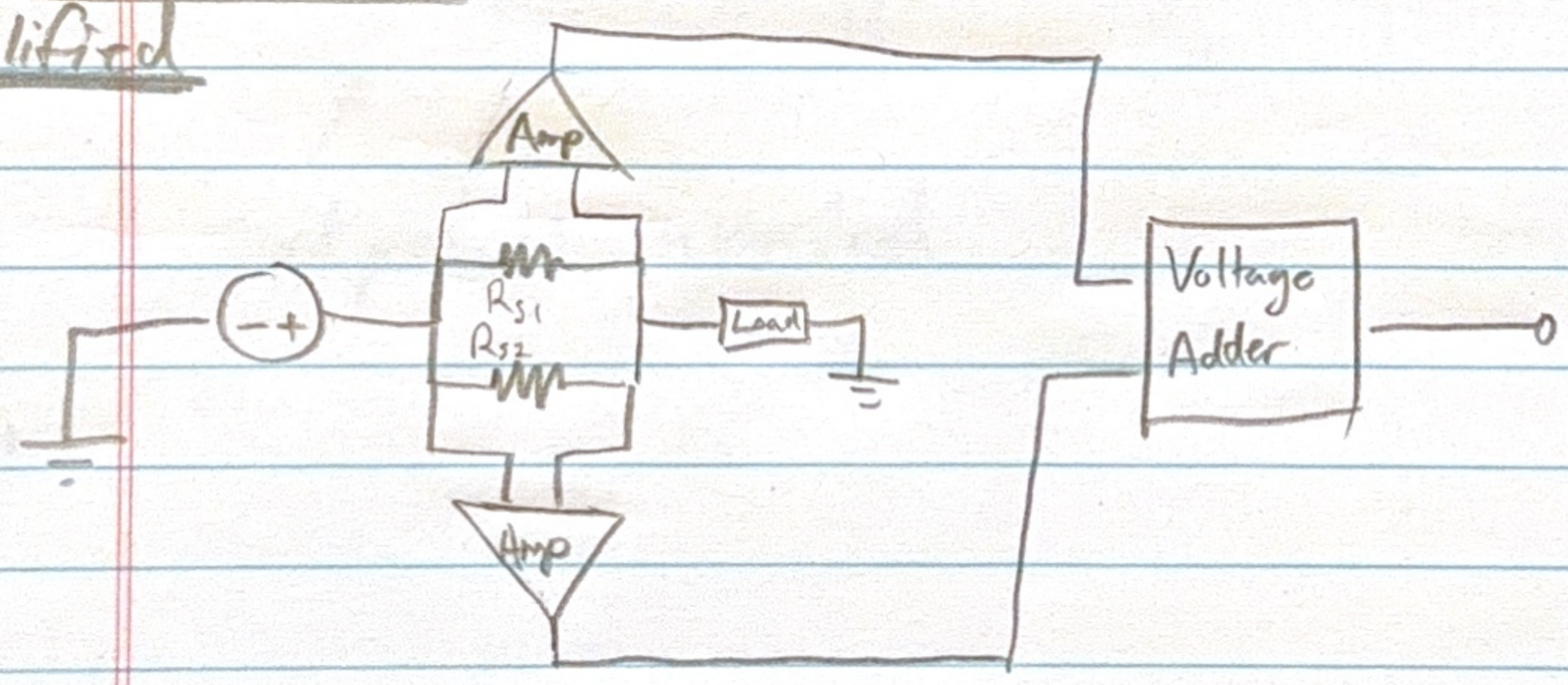


$I_{i1} + I_{i2} = I_i \rightarrow \frac{V_{S1}}{R_{S1}} + \frac{V_{S1}}{R_{S2}} = I_i$

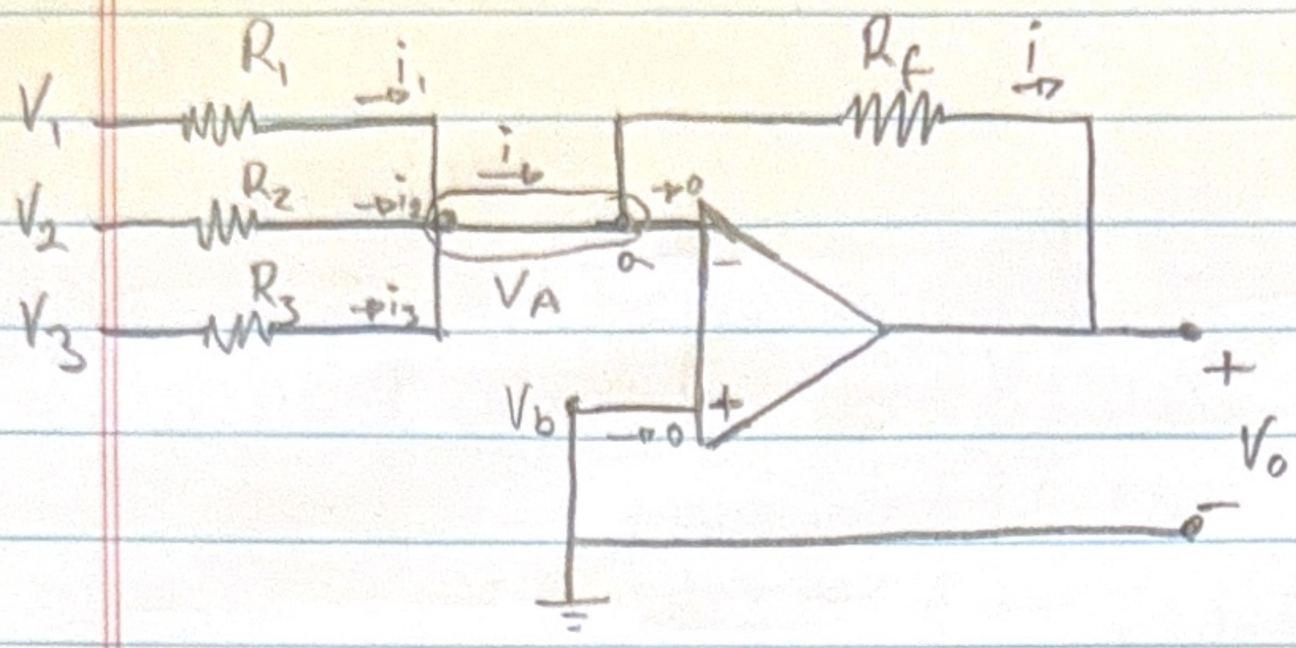
Dual shunt cont.



Simplified



Summing Amplifier



$i = i_1 + i_2 + i_3$, $i_1 = \frac{V_1 - V_a}{R_1}$, $i_2 = \frac{V_2 - V_a}{R_2}$, $i_3 = \frac{V_3 - V_a}{R_3}$

$$i = \frac{V_a - V_o}{R_f}$$

$$\frac{V_a - V_o}{R_f} = \frac{V_1 - V_a}{R_1} + \frac{V_2 - V_a}{R_2} + \frac{V_3 - V_a}{R_3}$$

For ideal op-amp $V_a = V_b = 0$

$$\frac{-V_o}{R_f} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

output voltage is the weighted sum of the inputs

↑ output is inverted